

Diagram

Figure 1: Diagram

Diagram

Figure 2: Diagram

01.2 — Schema Diagram

Audience: Anyone who needs a visual mental model of how Haritha's core DocTypes relate. **Goal:** ERD of the in-scope entities, with notes on which have custom fields and what they track. **Last updated:** 2026-08-29

Overview

This doc shows the **entity-relationship diagram** for Haritha Hospitals' in-scope modules: HR, Shift Management, Attendance, Leave, plus the supporting master data (Company, Department, Designation, Item, Customer, Supplier).

For the full customization catalog (CFs, PSs, etc.), see 01.1-schema-reference.md. For doc-level architecture (containers, network), see ../00-foundations/00.2-architecture-overview.md.

Main entities — ERD

Key relations — narrative

Foundation layer

- **Company** is the root — every master data DocType carries a company link. Haritha has 1 Company record: Haritha Hospitals.
- **Department** + **Designation** + **Employee Grade** are org-structure dimensions. An Employee belongs to exactly one of each.
- **Holiday List** is attached to Company + referenced by Shift Type + by Employee. Haritha has 1 Holiday List (Haritha Hospitals Holiday List) covering 14 Indian national holidays.

Employee master

- **Employee** is the central employee record. PK is autaname HR-EMP-NNNNN. The CSV source uses EMP-NNNN which maps via employee_number to the autaname.
- **Custom fields** on Employee include bank details (IFSC, MICR, account), tax (PAN), approvers (leave, expense, shift request), and HRMS-specific (health insurance, PF account, default shift, grade, employment type). See 01.1-schema-reference.md §Employee.
- **Attendance_device_id** (EMP-NNNN) is the FK join key from raw check-in logs to Employee.

Shift + Attendance workflow

- **Shift Type** defines a recurring shift pattern (start, end, grace periods, auto-attendance rules). Haritha has 25 Shift Types.
- **Shift Schedule Assignment** is the recurring template binding (employee × shift type, no date — recurring forever until cancelled).
- **Shift Assignment** is the dated instance (employee × shift type × specific date range). One SSA expands into many SAs as dates roll forward.
- **Employee Checkin** is raw IN/OUT from the attendance device.
- **Attendance** is derived from checkins (auto-attendance cron) OR manually marked. Status options: Present / Absent / On Leave / Half Day / Work From Home / **Holiday** / **Weekly Off** (last 2 added via Property Setter — see 01.1-schema-reference.md §Options Override).
- **Shift Request** is when an employee asks to swap a shift — separate workflow, not in current scope (Phase 2+).

Leave workflow

- **Leave Type** defines categories (Casual, Sick, Earned, etc.) with policies.
- **Leave Allocation** credits the employee with N days for a period.
- **Leave Application** is a request to consume allocation. Status: Open → Approved/Rejected.
- **Leave Block List** restricts when certain leave types can be applied (department-level).

Tax exemption workflow

- **Employee Tax Exemption Declaration** is the employee's projection (annual expected HRA, etc.).
- **Employee Tax Exemption Proof Submission** is the supporting documents (rent receipts, etc.) with actual amounts.
- Both have custom fields added by Haritha's HRMS install (HRA section, metro city flag, etc.) — see 01.1-schema-reference.md §Employee Tax Exemption Proof Submission / Declaration.

Payroll workflow

- **Salary Component** defines one line item (Basic, HRA, Provident Fund, ...). Custom field `component_type` tags it for payroll logic.
- **Salary Structure** assembles components into a formula (e.g., 40% Basic + 30% HRA + ...).
- **Salary Slip** is the per-period instance for one employee. Generated by Payroll Entry.

Master / commercial

- **Item Group + Item + UOM** — inventory basics. Haritha has 6 Item Groups + UoMs loaded.
- **Customer + Supplier** — AR/AP parties. Customer has `crm_deal` custom field for Frappe CRM integration.
- **Account + Cost Center** — Chart of Accounts. Haritha has 89 Accounts + 2 Cost Centers.

Diagram

Figure 3: Diagram

Custom field indicators

Below is a summary of **which entities have custom fields and what they track**. See 01.1-schema-reference.md for the full field-by-field catalog.

What the customization pattern reveals:

- **HR-heavy customization** — 7 of 8 multi-CF DocTypes are HRMS-owned (Employee, Department, Designation, both Tax Exemption forms). Company is the only non-HRMS DocType with double-digit CFs.
- **Approval workflows are pervasive** — Department has 3 approver tables; Employee has 3 approver Link fields. Haritha routes approvals by role × department.
- **Tax/PF/HRA compliance** — the bulk of HR CFs relate to Indian statutory compliance (PAN, IFSC, PF account, HRA declaration/proof).
- **Stock commercial is mostly untouched** — Sales/Purchase/Inventory DocTypes appear only in the Property Setter list (mostly hidden cleanups), not in the Custom Field list.

How to render this diagram

On GitHub

GitHub renders Mermaid automatically when you put the mermaid block inside a Markdown file. Just open the doc on GitHub and the diagrams show up inline. No setup needed.

Offline

Install Mermaid CLI:

```
npm install -g @mermaid-js/mermaid-cli
# or
brew install mermaid-cli
```

Render a single file:

```
mmdc -i 01.2-schema-diagram.md -o 01.2-schema-diagram.png -w 2400
```

The -w 2400 flag gives a wide image to fit the ERD. PNG export needs the Mermaid CLI + a headless Chromium.

Inside Frappe (alternative)

If you want the ERD rendered inside Frappe itself, paste the mermaid block into a Wiki page or a DocType description (Markdown is rendered in those). The Desk uses a Markdown renderer that includes Mermaid support.

Excalidraw / draw.io alternative

If you need to edit the diagram (e.g., add a new DocType), the Mermaid syntax above can be pasted into:

- Mermaid Live Editor (free, instant render)
- VS Code with the “Markdown Preview Mermaid Support” extension
- Excalidraw (after conversion to Excalidraw syntax via mermaid-to-excalidraw)

We prefer Mermaid because it's **text-based → git diffable → reviewable in PRs**.

What this doc doesn't cover

Topic	Doc
All 274 customizations catalog (every field's purpose)	01.1-schema-reference.md
Shift Management workflow (Roster, Auto Attendance, Manual Attendance, Leave)	docs/phase6/02-shift-workflow/* (Batch 2)
Operational data — current values for each entity	docs/HARITHA_HOSPITALS_GUIDE.md Appendix A
Migration playbook — how to set up these tables from CSV	docs/HARITHA_HOSPITALS_GUIDE.md §6

End of 01.2. Next: Tier 2 — Shift Workflow (Batch 2).